6) How many 32 K X 1 RAM chips are needed to provide a memory capacity of 256 K-bytes

a 8
b 32
c 64
d 128

RAM chip's capacity $= 32K \times 1$

$= 32 \times 1024 \times 1$ bits

Memory capacity $= 256$ K bytes

$= 256 \times 1024 \times 8$ bits

No. of chips needed $= \dfrac{\text{Memory capacity}}{\text{RAM chip's capacity}}$

$$= \frac{256 \times 1024 \times 8}{32 \times 1024 \times 1}$$

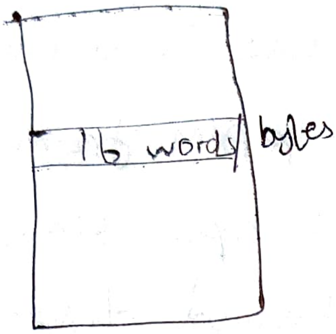
$$= \frac{2^8 \times 2^{10} \times 2^3}{2^5 \times 2^{10} \times 1}$$

$$= 2^3 \times 2^3$$

$$= 2^6$$

$$= 64 \text{ chips}$$

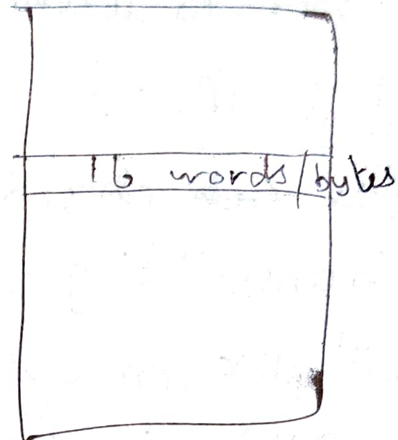
7) For a memory system having the following specification: size of the main memory is 4K blocks, size of the cache is 128 blocks and the block size is 16 words. Assuming that the system uses associative mapping, the cache field parameters would be:



cache

128 blocks of
16 words

ie 128 x 16



main mly

4 K blocks of 16 words

ie 4K x 16

Tag 16 bit word

Tag/Block no:	Block offset
16-4 = 12 bits	→ 16 bytes = 2^4 = 4 bits

Size of main mly = 4K x 16 ~~bytes~~

$= 4 \times 1024 \times 16$

$= 2^2 \times 2^{10} \times 2^4$

$= 2^{16}$ bytes = 16 bit

∴ Answer is

word field = 4 bits, Tag field = 12 bits, No: of bits in main memory address = 16 bits.

8) A processor is fetching instructions at the rate of 1 MIPS; a DMA module is used to transfer ~~each~~ characters to RAM from a device transmitting at 9600 bps. How much time will the processor be slowed down due to DMA activity?

a. 9.6 ms

b. 4.8 ms

c. 2.4 ms

d. 1.2 ms

External device sends data to DMA module in 1 sec = 9600 bits per second

$= \frac{9600}{8}$

= 1200 bytes per second

1 MIPS → means 1 million instructions per second

processor runs instructions at the rate of

= 1 Million per sec → 10,00,000 · 10 lakh

$= 10^6$

∴ The time will the processor be slowed down due to DMA activity

$$= \frac{1200}{10^6}$$

$= 0.0012$ seconds

1000ms = 1sec

$= 1.2$ ms

9. A pipeline P operating at 400 MHz has a speedup factor of 6 and operating at 70% efficiency. How many stages are there in the pipeline?

a. 5
b. 6
c. 8
d. 9

$$\text{Efficiency} = \frac{\text{Speedup factor}}{\text{Number of stages}}$$

$$\frac{70}{100} = \frac{6}{\text{No: of stage}}$$

$$\text{No: of stages} = \frac{6 \times 100}{70}$$

$$\begin{array}{r} 8.56 \\ 7 \overline{) 60} \\ 56 \\ \hline 40 \\ 35 \\ \hline 50 \\ 42 \end{array}$$

$= 8.56$

$= \underline{\underline{9}}$

10. The number of address and data lines for a memory of 4K x 16 is

Ans: C

a) 10 and 16
b. 11 and 8
c. 12 and 16
d. 12 and 12

Ans. ROM memory size $= 2^m \times n$

$m =$ no: of address lines

$n =$ no: of data line

Given $4K \times 16$

$= 2^2 \times 2^{10} \times 16$

$= 2^{12} \times 16 \Rightarrow (2^m \times n)$

$\therefore$ No: of address line $= 12$

No: of data line $= 16$

11. Unix, DOS, DR FlexOS, IBM OS/2, Microsoft windows, ReactOS – mkdir

DOS, OS/2, windows, ReactOS – md

both means make directory

12. The stage delays in a 4-stage pipeline are 800, 500, 400 and 300 picoseconds. The first stage (with delay 800 picoseconds) is replaced with a functionally equivalent design involving two stages with respective delays 600 and 350 picoseconds. The throughput increase of the pipeline is percent.

Ans: Throughput of the pipeline = $\dfrac{\text{Number of instructions}}{\text{Total time to complete the instructions}}$

Throughput = no: of instructions executed per unit time

$= \dfrac{1}{T_p}$

Initial stage delays:

800, 500, 400 & 300 picoseconds

No: of stages = ~~4~~ 4

In case of pipeline, CPI = 1

[The pipelined processor ideally would have a ~~Gp~~ Cycle per Instruction of 1]

$\therefore$ Throughput = $\dfrac{1}{800}$

Updated stage delay:

600, 350, 500, 400, 300 picoseconds

Max stage delay = 600 picoseconds

Throughput in this case = $\dfrac{1}{600}$

So percentage increase in throughput

$= \left(\dfrac{\text{New} - \text{Old}}{\text{Old}}\right) \times 100$

$$= \frac{\frac{1}{600} - \frac{1}{800}}{\frac{1}{800}}$$

$= 33.33\%$

ISP-Instruction set Processor

______ is used to store data in registers

a D flip flop

b. JK flip flop

c. RS flip flop

d) NOTA

Ans: a